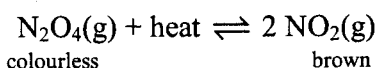


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255
255

- A.** The rate of the forward reaction is equal to the rate of the reverse reaction.
B. The concentration of the reactants equals the concentration of the products.
C. The rate of the forward reaction is greater than the rate of the reverse reaction.
D. The concentration of the products is greater than the concentration of the reactants.

Use the following information to answer the next question.



2. Which of the following properties would **not** be used to determine if the equilibrium system represented by the equation above is at equilibrium?
- A. mass
- B. colour
- C. pressure
- D. temperature

Use the following information to answer the next question.

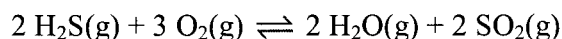
For the steam-hydrocarbon reforming process, the equilibrium law expression is

$$K_c = \frac{[\text{CO}_2(\text{g})][\text{H}_2(\text{g})]^4}{[\text{CH}_4(\text{g})][\text{H}_2\text{O}(\text{g})]^2}$$

3. The reaction described by this equilibrium is
- A. $\text{CH}_4(\text{g}) + 2 \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + 4 \text{H}_2(\text{g})$
- B. $\text{CO}_2(\text{g}) + 4 \text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$
- C. $\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g})$
- D. $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g})$

Use the following information to answer the next question.

Coal and natural gas contain trace amounts of sulfur compounds which, when burned, may lead to acid rain pollution. A reaction that is related to acid rain is



4. The K_c expression for this reaction is

A. $K_c = \frac{[\text{H}_2\text{O}(\text{g})]^2 + [\text{SO}_2(\text{g})]^2}{[\text{H}_2\text{S}(\text{g})]^2 + [\text{O}_2(\text{g})]^3}$

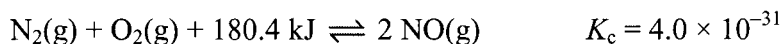
B. $K_c = \frac{[\text{H}_2\text{S}(\text{g})]^2 + [\text{O}_2(\text{g})]^3}{[\text{H}_2\text{O}(\text{g})]^2 + [\text{SO}_2(\text{g})]^2}$

C. $K_c = \frac{[\text{H}_2\text{O}(\text{g})]^2 [\text{SO}_2(\text{g})]^2}{[\text{H}_2\text{S}(\text{g})]^2 [\text{O}_2(\text{g})]^3}$

D. $K_c = \frac{[\text{H}_2\text{S}(\text{g})]^2 [\text{O}_2(\text{g})]^3}{[\text{H}_2\text{O}(\text{g})]^2 [\text{SO}_2(\text{g})]^2}$

Use the following information to answer the next question.

Nitrogen fixation occurs slowly in the atmosphere. The equation for this reaction is



5. At equilibrium, if $[\text{O}_2(\text{g})] = [\text{N}_2(\text{g})]$, then

A. $[\text{NO}(\text{g})] = [\text{N}_2(\text{g})]$

B. $[\text{NO}(\text{g})] > [\text{N}_2(\text{g})]$

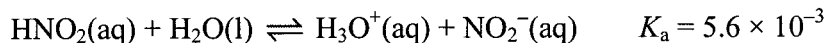
C. $[\text{NO}(\text{g})] = 2 [\text{N}_2(\text{g})]$

D. $[\text{NO}(\text{g})] < [\text{N}_2(\text{g})]$

Numerical Response

6. At a temperature of 300°C and a pressure of 40.5 kPa, 90.0 mol of H₂(g) and 80.0 mol of N₂(g) are injected into a 1.00 L reaction vessel. When equilibrium is established, 20.0 mol of NH₃(g) are present. The chemical amount of H₂(g) present in this equilibrium mixture is 60 mol.

Use the following information to answer the next question.



7. The equilibrium law expression for the equation above is i, and at equilibrium the amount of products present is ii the amount of reactants present.

The statement above is completed by the information in row

Row	i	ii
A.	$K_a = \frac{[\text{H}_3\text{O}^+(\text{aq})][\text{NO}_2^-(\text{aq})]}{[\text{HNO}_2(\text{aq})]}$	less than
B.	$K_a = \frac{[\text{H}_3\text{O}^+(\text{aq})][\text{NO}_2^-(\text{aq})]}{[\text{HNO}_2(\text{aq})][\text{H}_2\text{O}(\text{l})]}$	greater than
C.	$K_a = \frac{[\text{HNO}_2(\text{aq})][\text{H}_2\text{O}(\text{l})]}{[\text{H}_3\text{O}^+(\text{aq})][\text{NO}_2^-(\text{aq})]}$	less than
D.	$K_a = \frac{[\text{HNO}_2(\text{aq})]}{[\text{H}_3\text{O}^+(\text{aq})][\text{NO}_2^-(\text{aq})]}$	greater than

Use the following information to answer the next question.

$$K_c = \frac{[\text{NO}_2(\text{g})]^2}{[\text{NO}(\text{g})]^2 [\text{O}_2(\text{g})]}$$

8. The equation that is represented by the above equilibrium law expression is

- A. $2 \text{NO}_2(\text{g}) \rightleftharpoons 2 \text{NO}(\text{g}) + \text{O}_2(\text{g})$
- B. $2 \text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{NO}_2(\text{g})$
- C. $\text{NO}_2(\text{g}) \rightleftharpoons \text{NO}(\text{g}) + \text{O}_2(\text{g})$
- D. $\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons \text{NO}_2(\text{g})$

Use the following information to answer the next question.

The equilibrium law expression for an industrial method of producing ethanol is shown below.

$$K_c = \frac{[\text{C}_2\text{H}_5\text{OH}(\text{g})]}{[\text{C}_2\text{H}_4(\text{g})][\text{H}_2\text{O}(\text{g})]}$$

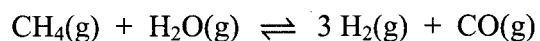
Under certain conditions, $K_c = 300.0$. At equilibrium, a 5 000 L reaction vessel contains 115 mol of $\text{C}_2\text{H}_4(\text{g})$ and 110 mol of $\text{H}_2\text{O}(\text{g})$.

9. Under these conditions, the equilibrium concentration of $\text{C}_2\text{H}_5\text{OH}(\text{g})$ is

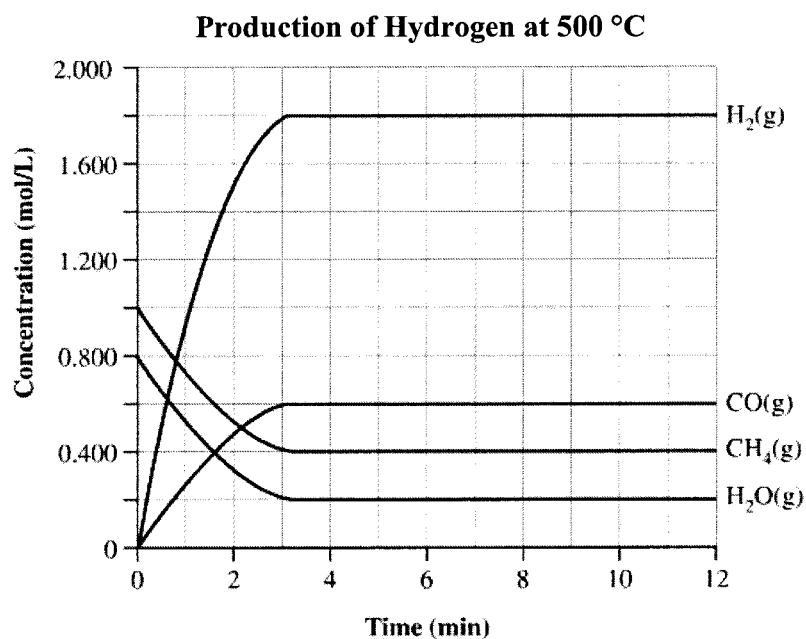
- A. $1.60 \times 10^{-6} \text{ mol/L}$
- B. $1.52 \times 10^{-1} \text{ mol/L}$
- C. $7.50 \times 10^1 \text{ mol/L}$
- D. $5.92 \times 10^5 \text{ mol/L}$

Use the following information to answer the next question.

A technician is producing hydrogen gas. He adds methane gas, steam, and a nickel catalyst to an empty reaction container and allows the system to reach equilibrium. The reaction is represented by the following equation.



The technician's data are represented by the following graph.

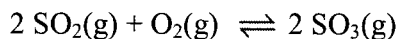


Numerical Response

10. The equilibrium constant for the reaction represented in the graph above is _____.

Use the following information to answer the next **two** questions.

When the system represented by the equation below is at equilibrium in a 2.00 L flask at 15.0 °C, the flask contains 1.15 mol of SO₂(g), 2.13 mol of O₂(g), and 3.74 mol of SO₃(g).



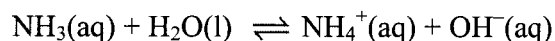
11. Which of the following changes, when applied to the equilibrium system above, would change the value of the equilibrium constant?
- A. The addition of a catalyst
 - B. An increase in temperature
 - C. The addition of hydrogen gas
 - D. A decrease in the volume of the container

Numerical Response

12. At 15.0 °C, the equilibrium constant is _____.

Use the following information to answer the next question.

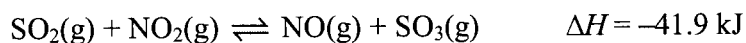
The reaction of ammonia in water is represented by the following equilibrium equation.



13. The addition of NaOH(aq) to the equilibrium above would cause the
- A. equilibrium constant to decrease
 - B. equilibrium constant to increase
 - C. concentration of NH₃(aq) to decrease
 - D. concentration of NH₄⁺(aq) to decrease

Use the following information to answer the next question.

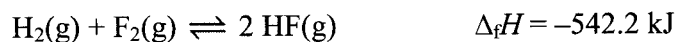
Some of the $\text{SO}_2(\text{g})$ produced from the burning of coal and natural gas can react with $\text{NO}_2(\text{g})$ in the atmosphere according to the following equation.



14. The equilibrium concentration of $\text{SO}_3(\text{g})$ in the above reaction could be increased by
- A. adding $\text{NO}_2(\text{g})$
 - B. adding a catalyst
 - C. removing $\text{SO}_2(\text{g})$
 - D. increasing the temperature of the reaction vessel

Use the following information to answer the next question.

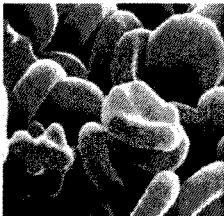
Hydrogen fluoride is produced by reacting hydrogen with fluorine according to the following equation.



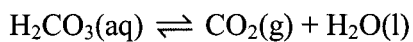
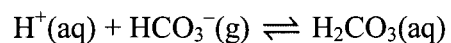
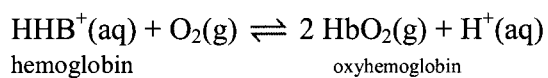
15. A stress that would shift the equilibrium towards the products would be to
- A. add $\text{HF}(\text{g})$
 - B. remove $\text{H}_2(\text{g})$
 - C. decrease the volume of the reaction vessel
 - D. decrease the temperature of the reaction vessel

Use the following information to answer the next question.

Red Blood Cells



Three Important Equilibria in Blood

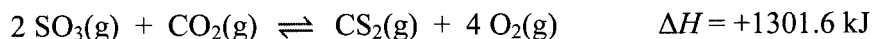


16. In blood, the $[\text{H}^+(\text{aq})]$ could be increased by decreasing the

- A. $[\text{O}_2(\text{g})]$
- B. $[\text{CO}_2(\text{g})]$
- C. $[\text{HCO}_3^-(\text{aq})]$
- D. $[\text{H}_2\text{CO}_3(\text{aq})]$

Use the following information to answer the next **two** questions.

A technician places a sample of sulfur trioxide gas and a sample of carbon dioxide gas in an empty 1.0 L flask and allows them to reach equilibrium at 800 °C. The equilibrium system that is established in the flask is represented by the following equation.



Changes to the Equilibrium System

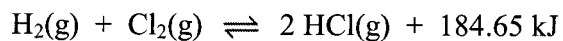
- I** The addition of a catalyst
- II** A decrease in the volume of the container
- III** An increase in the temperature of the system

Numerical Response

- 17.** If at equilibrium the concentration of $\text{SO}_3(\text{g})$ is 0.289 mol/L; the concentration of $\text{CO}_2(\text{g})$ is 0.0180 mol/L; the concentration of $\text{CS}_2(\text{g})$ is 0.201 mol/L; and the concentration of $\text{O}_2(\text{g})$ is 0.900 mol/L, then the K_c value of the system at 800 °C is _____ .
- 18.** Which of the changes to the equilibrium system numbered above would affect the value of the equilibrium constant?
- A.** I and III only
 - B.** I, II, and III
 - C.** II only
 - D.** III only

Use the following information to answer the next question.

Equilibrium System



Stresses

- I** The addition of $\text{H}_2(\text{g})$
- II** The addition of $\text{HCl}(\text{g})$
- III** The cooling of the equilibrium system
- IV** An increase in the volume of the container

19. The stresses numbered above that would cause the equilibrium system to shift toward the products are
- A. I and III only
 - B. I, III, and IV
 - C. II and III only
 - D. II, III, and IV

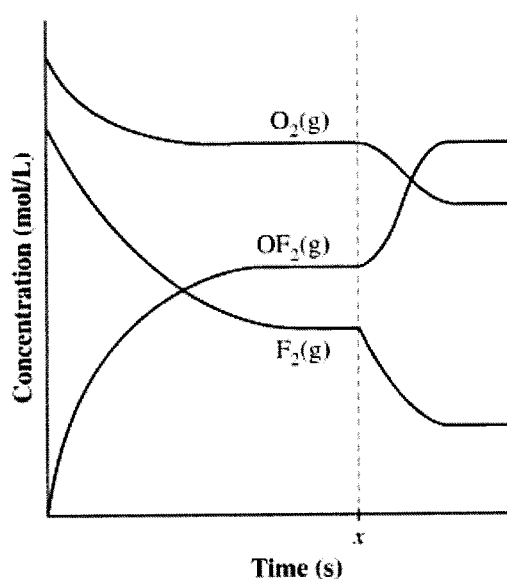
Use the following information to answer the next question.

A technician injected fluorine gas and oxygen gas into an empty 1.0 L reaction vessel. She closed the vessel and allowed the reaction to reach equilibrium, as represented by the following equation.



The technician changed the reaction conditions and allowed a new equilibrium to be established as represented by the graph below.

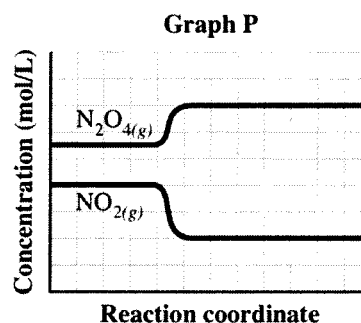
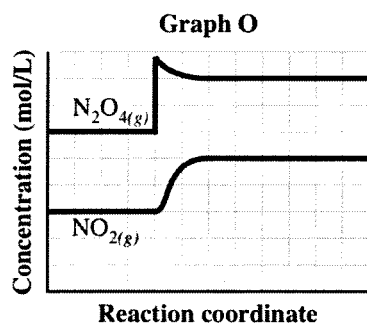
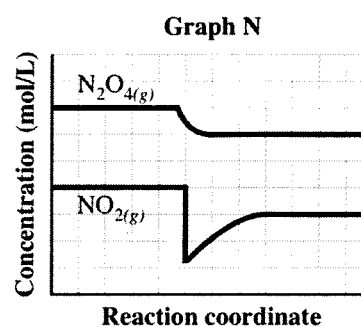
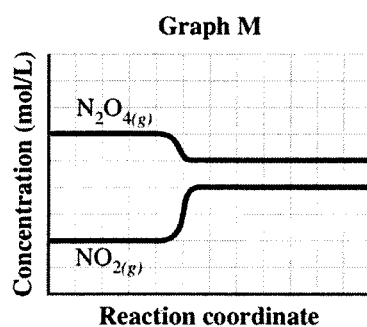
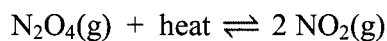
Changes to the Equilibrium System



20. Which of the following stresses represents the change in the equilibrium system at time x on the graph above?
- A. A catalyst is added.
 - B. Pressure is decreased.
 - C. Temperature is increased.
 - D. Fluorine gas is removed from the vessel.

Use the following information to answer the next question.

The graphs below show the relative concentration of $\text{N}_2\text{O}_4(\text{g})$ and $\text{NO}_2(\text{g})$ as various stresses are exerted on the equilibrium system.



- Stresses:**
- | | |
|------------------------------------|---|
| 1 removing $\text{NO}_2(\text{g})$ | 3 adding heat |
| 2 cooling system | 4 adding $\text{N}_2\text{O}_4(\text{g})$ |

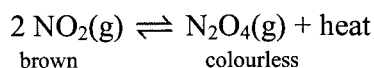
Numerical Response

21. Match each of the graphs with the appropriate stress.

Graph M _____ (Record in **column 1**)
 Graph N _____ (Record in **column 2**)
 Graph O _____ (Record in **column 3**)
 Graph P _____ (Record in **column 4**)

22. When applied to an equilibrium system, which of the following stresses would cause a change in the K_c value after the equilibrium has been re-established?
- A. addition of a catalyst
 - B. decrease in temperature by cooling the system
 - C. addition of an inert gas to increase the pressure
 - D. decrease in concentration by removing a product

Use the following information to answer the next question.



Stresses Applied to an Equilibrium System

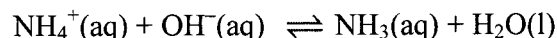
- | | |
|------------------------|--|
| 1 increase volume | 5 increase $\text{NO}_2(\text{g})$ concentration |
| 2 decrease volume | 6 decrease $\text{NO}_2(\text{g})$ concentration |
| 3 increase temperature | 7 increase $\text{N}_2\text{O}_4(\text{aq})$ concentration |
| 4 decrease temperature | 8 decrease $\text{N}_2\text{O}_4(\text{aq})$ concentration |

Numerical Response

23. The stresses numbered above that will cause the equilibrium system to shift to the left are _____, _____, and _____.

Use the following information to answer the next question.

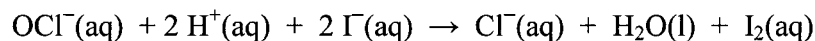
Two cleaning solutions were accidentally mixed. A strong smell of ammonia alerted a technician to the accident. After checking the labels of the cleaners and discovering that one container held $\text{NH}_4\text{Cl}(\text{aq})$, and the other held $\text{KOH}(\text{aq})$, the technician determined that the smell came from the following reaction.



24. In this equilibrium, the Brønsted-Lowry acids are
- A. $\text{NH}_3(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$
 - B. $\text{NH}_4^+(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$
 - C. $\text{NH}_3(\text{aq})$ and $\text{OH}^-(\text{aq})$
 - D. $\text{NH}_4^+(\text{aq})$ and $\text{OH}^-(\text{aq})$

Use the following information to answer the next question.

The concentration of aqueous hypochlorite in laundry bleach, NaOCl(aq) , can be determined by titrating a sample with an iodide solution, as represented by the following equation.



25. If during the titration the pH of the solution changes from 9.2 to 7.2, then the acidity of the solution i and the hydronium ion concentration ii by a factor of 100.

The statement above is completed by the information in row

Row	i	ii
A.	decreases	increases
B.	decreases	decreases
C.	increases	increases
D.	increases	decreases

Use the following information to answer the next question.

A solution containing the ammonium ion, $\text{NH}_4^+(\text{aq})$, is mixed with a solution containing the hydrogen carbonate ion, $\text{HCO}_3^-(\text{aq})$.

26. In the forward reaction, the Brønsted-Lowry base is i and its conjugate acid is ii.

The statement above is completed by the information in row

Row	i	ii
A.	$\text{HCO}_3^-(\text{aq})$	$\text{H}_2\text{CO}_3(\text{aq})$
B.	$\text{HCO}_3^-(\text{aq})$	$\text{NH}_4^+(\text{aq})$
C.	$\text{NH}_4^+(\text{aq})$	$\text{HCO}_3^-(\text{aq})$
D.	$\text{NH}_4^+(\text{aq})$	$\text{NH}_3(\text{aq})$

Use the following information to answer the next question.

A sample of hydrogen iodide gas, HI(g), is collected and then dissolved in 1.00 L of water.

Statements About Solutions

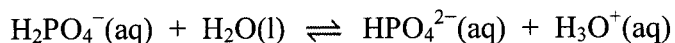
- 1 The pH is less than the pOH.
- 2 The pH is greater than the pOH.
- 3 The solution conducts electricity.
- 4 The solution does not conduct electricity.
- 5 Almost all of the HI(aq) ionizes in the solution.
- 6 Most of the HI(aq) does not ionize in the solution.
- 7 The concentration of $\text{H}_3\text{O}^+(\text{aq})$ is less than the concentration of $\text{OH}^-(\text{aq})$.
- 8 The concentration of $\text{H}_3\text{O}^+(\text{aq})$ is greater than the concentration of $\text{OH}^-(\text{aq})$.

Numerical Response

27. The statements above that describe the hydrogen iodide solution are numbered _____, _____, _____, and _____.

Use the following information to answer the next question.

The following equilibrium system maintains a constant pH in the intracellular fluid in the body.



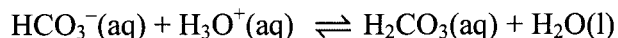
28. A Brønsted-Lowry acid in the equilibrium system represented by the equation above is i , and it will ii the Brønsted-Lowry base.

The statement above is completed by the information in row

Row	i	ii
A.	$\text{H}_2\text{O}(\text{l})$	donate protons to
B.	$\text{H}_2\text{O}(\text{l})$	accept protons from
C.	$\text{H}_2\text{PO}_4^-(\text{aq})$	donate protons to
D.	$\text{H}_2\text{PO}_4^-(\text{aq})$	accept protons from

Use the following information to answer the next question.

The reaction that occurs when HCl(aq) is added to a hydrogen carbonate ion solution is



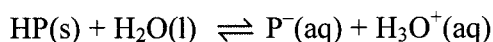
29. The amphiprotic species in the system above are

- A. $\text{HCO}_3^-(\text{aq})$ and $\text{H}_3\text{O}^+(\text{aq})$
- B. $\text{H}_3\text{O}^+(\text{aq})$ and $\text{H}_2\text{CO}_3(\text{aq})$
- C. $\text{HCO}_3^-(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$
- D. $\text{H}_3\text{O}^+(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$

Use the following information to answer the next **two** questions.

Antibiotics that are formed by different species of *Penicillium* bacteria are among the most widely prescribed drugs in the world today.

One of these antibiotics is penicillin G (benzylpenicillinic acid), which is represented by the formula HP(s) . This antibiotic is only partially soluble in water. At body temperature, this saturated solution contains $1.20 \times 10^{-2} \text{ mol/L}$ benzylpenicillinic acid and has a pH of 2.420. The ionization of this acid is represented by the following equilibrium equation.



30. Equilibrium occurs when the rate of the forward reaction is

- A. equal to the rate of the reverse reaction
- B. faster than the rate of the reverse reaction
- C. slower than the rate of the reverse reaction
- D. favoured over the rate of the reverse reaction

31. In terms of strength and K_a , penicillin G is a

- A. strong acid and has a $K_a > 1$
- B. strong acid and has a $K_a < 1$
- C. weak acid and has a $K_a > 1$
- D. weak acid and has a $K_a < 1$

Use the following information to answer the next question.

Dermatologists use acids such as glycolic acid, $\text{CH}_2\text{OHCOOH}(\text{aq})$, to remove the surface layer of skin, revealing smoother skin underneath. In high concentrations, glycolic acid can remove precancerous lesions or thickened skin.

K_a Values of Acids Used by Dermatologists

	Monoprotic Acids	K_a Values
1	Salicylic acid	1.0×10^{-3}
2	Glycolic acid	1.5×10^{-4}
3	Lactic acid	1.4×10^{-4}
4	Trichloroacetic acid	3.0×10^{-1}

Numerical Response

32. The acids above, listed from **strongest** to **weakest**, are _____, _____, _____, and _____.
Strongest Weakest

Use the following information to answer the next question.

Reagents

1 $\text{Na}_2\text{CO}_3(\text{aq})$	5 $\text{Na}_2\text{HPO}_4(\text{aq})$
2 $\text{CaO}(\text{aq})$	6 $\text{CH}_3\text{COOH}(\text{aq})$
3 $\text{NaHCO}_3(\text{aq})$	7 $\text{KOOCCOOH}(\text{aq})$
4 $\text{Ba}(\text{OH})_2(\text{aq})$	8 $\text{NaHSO}_3(\text{aq})$

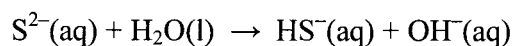
Numerical Response

33. Listed in numerical order, the reagents above that are amphoteric are _____, _____, _____, and _____.

Use the following information to answer the next question.

Body hair can be removed using a chemical hair-removal cream. One form of the cream contains sodium sulfide, calcium sulfide, and strontium sulfide. The sulfide ions in the cream will react water, as represented by the following equation.

Hair-Removal Reaction



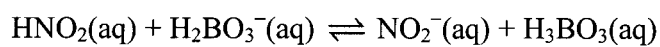
The solution resulting from the reaction will break some of the bonds in the hair, which allows the hair to be washed away.

34. In the hair-removal reaction, the sulfide ion acts as a Brønsted-Lowry i , and ii a proton.

The statement above is completed by the information in row

Row	i	ii
A.	base	donates
B.	base	accepts
C.	acid	donates
D.	acid	accepts

Use the following information to answer the next question.



35. The amphiprotic species in the above reaction is
- A. $\text{H}_2\text{BO}_3^{-}(\text{aq})$
 - B. $\text{HNO}_2(\text{aq})$
 - C. $\text{NO}_2^{-}(\text{aq})$
 - D. $\text{H}_3\text{BO}_3(\text{aq})$

36. The Brønsted–Lowry equation that best represents the equilibrium in an aqueous solution of ammonia is

- A. $\text{NH}_3(\text{aq}) \rightleftharpoons 3 \text{H}^+(\text{aq}) + \text{N}^{3-}(\text{aq})$
- B. $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4\text{OH}(\text{aq})$
- C. $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
- D. $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{NH}_2^-(\text{aq})$

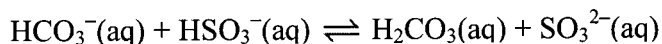
37. The equation that represents the equilibrium of phosphoric acid is

- A. $\text{H}_3\text{PO}_4(\text{aq}) \rightleftharpoons 3 \text{H}^+(\text{aq}) + \text{PO}_4^{3-}(\text{aq})$
- B. $\text{H}_3\text{PO}_4(\text{aq}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{PO}_3^{3-}(\text{aq})$
- C. $\text{H}_3\text{PO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{PO}_4^-(\text{aq}) + \text{OH}^-(\text{aq})$
- D. $\text{H}_3\text{PO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{H}_2\text{PO}_4^-(\text{aq})$

38. The substance that can act as a Brønsted–Lowry base is

- A. $\text{NaCl}(\text{aq})$
- B. $\text{CH}_3\text{OH}(\text{aq})$
- C. $\text{HCOOH}(\text{aq})$
- D. $\text{NaHCO}_3(\text{aq})$

Use the following information to answer the next question.



39. The Brønsted–Lowry acids in the above equation are

- A. $\text{HCO}_3^-(\text{aq})$ and $\text{HSO}_3^-(\text{aq})$
- B. $\text{HCO}_3^-(\text{aq})$ and $\text{H}_2\text{CO}_3(\text{aq})$
- C. $\text{HSO}_3^-(\text{aq})$ and $\text{H}_2\text{CO}_3(\text{aq})$
- D. $\text{HSO}_3^-(\text{aq})$ and $\text{SO}_3^{2-}(\text{aq})$

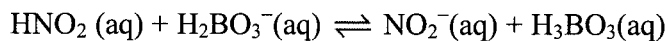
40. Which of the following equations represents a weak base system at equilibrium?

- A. $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
- B. $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
- C. $\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
- D. $\text{HNO}_2(\text{aq}) + \text{CN}^-(\text{aq}) \rightleftharpoons \text{NO}_2^-(\text{aq}) + \text{HCN}(\text{aq})$

41. Which of the following substances is an amphiprotic species?

- A. $\text{CH}_4(\text{g})$
- B. $\text{CH}_3\text{OH}(\text{aq})$
- C. $\text{CH}_3\text{COO}^-(\text{aq})$
- D. $\text{HOCCOO}^-(\text{aq})$

Use the following information to answer the next question.



42. In the reaction represented by the equation above, an amphiprotic species is i , and a conjugate acid-base pair is ii .

The statement above is completed by the information in row

Row	i	ii
A.	$\text{HNO}_2(\text{aq})$	$\text{HNO}_2(\text{aq})$ and $\text{NO}_2^-(\text{aq})$
B.	$\text{HNO}_2(\text{aq})$	$\text{H}_2\text{BO}_3^-(\text{aq})$ and $\text{NO}_2^-(\text{aq})$
C.	$\text{H}_2\text{BO}_3^-(\text{aq})$	$\text{HNO}_2(\text{aq})$ and $\text{NO}_2^-(\text{aq})$
D.	$\text{H}_2\text{BO}_3^-(\text{aq})$	$\text{H}_2\text{BO}_3^-(\text{aq})$ and $\text{NO}_2^-(\text{aq})$

43. The conjugate base of $\text{N}_2\text{H}_5^+(\text{aq})$ is

- A. $\text{H}_2\text{O}(\text{l})$
- B. $\text{OH}^-(\text{aq})$
- C. $\text{N}_2\text{H}_4(\text{aq})$
- D. $\text{N}_2\text{H}_6^{2+}(\text{aq})$

Use the following information to answer the next question.

Stinging ants are very common in the Amazon basin. These ants produce methanoic acid in such large quantities that it becomes a pollutant in streams. Methanoic acid dissociates in water, as represented by the following equilibrium equation.

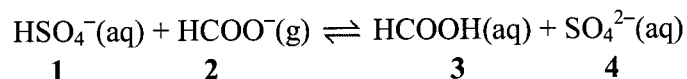


44. A conjugate acid-base pair in the equilibrium equation above is

- A. HCOOH(aq) and $\text{H}_3\text{O}^+\text{(aq)}$
- B. HCOOH(aq) and $\text{H}_2\text{O(l)}$
- C. $\text{H}_2\text{O(l)}$ and $\text{HCOO}^-\text{(aq)}$
- D. $\text{H}_2\text{O(l)}$ and $\text{H}_3\text{O}^+\text{(aq)}$

Use the following information to answer the next question.

Nitrogen fixation occurs slowly in the atmosphere. The equation for this reaction is



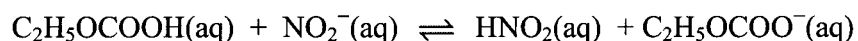
Numerical Response

45. For the above reaction, the acid and its conjugate base and then the base and its conjugate acid, listed in that order are _____, _____, _____, and _____.

46. In which of the following reactions does equilibrium favour the products?

- A. $\text{HSO}_4^-\text{(aq)} + \text{F}^-\text{(aq)} \rightleftharpoons \text{HF(aq)} + \text{SO}_4^{2-}\text{(aq)}$
- B. $\text{HF(aq)} + \text{H}_2\text{O(l)} \rightleftharpoons \text{H}_3\text{O}^+\text{(aq)} + \text{F}^-\text{(aq)}$
- C. $\text{HF(aq)} + \text{SO}_4^{2-}\text{(aq)} \rightleftharpoons \text{HSO}_4^-\text{(aq)} + \text{F}^-\text{(aq)}$
- D. $\text{CH}_3\text{COOH(aq)} + \text{F}^-\text{(aq)} \rightleftharpoons \text{HF(aq)} + \text{CH}_3\text{COO}^-\text{(aq)}$

Use the following information to answer the next question.



47. Which of the following rows identifies the stronger acid and the side of the equilibrium that is favoured?

The statement above is completed by the information in row

Row	Stronger Acid	Side Favoured
A.	$\text{HNO}_2(\text{aq})$	products
B.	$\text{HNO}_2(\text{aq})$	reactants
C.	$\text{C}_2\text{H}_5\text{OCOOH}(\text{aq})$	products
D.	$\text{C}_2\text{H}_5\text{OCOOH}(\text{aq})$	reactants

Use the following information to answer the next question.

Monoprotic Organic Acids			
Acid Name	Acid Formula	Conjugate Base Formula	K_a
Chloroacetic acid	$\text{C}_2\text{H}_3\text{ClO}_2(\text{aq})$	$\text{C}_2\text{H}_2\text{ClO}_2^-(\text{aq})$	1.3×10^{-3}
Salicylic acid	$\text{C}_7\text{H}_6\text{O}_3(\text{aq})$	$\text{C}_7\text{H}_5\text{O}_3^-(\text{aq})$	1.0×10^{-3}
Glycolic acid	$\text{C}_2\text{H}_4\text{O}_3(\text{aq})$	$\text{C}_2\text{H}_3\text{O}_3^-(\text{aq})$	1.5×10^{-4}
Lactic acid	$\text{C}_3\text{H}_6\text{O}_3(\text{aq})$	$\text{C}_3\text{H}_5\text{O}_3^-(\text{aq})$	1.4×10^{-4}

48. The Brønsted-Lowry equation that represents a reaction that favours the products is

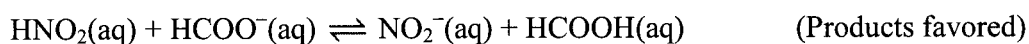
- A. $\text{C}_3\text{H}_6\text{O}_3(\text{aq}) + \text{C}_2\text{H}_3\text{O}_3^-(\text{aq}) \rightleftharpoons \text{C}_3\text{H}_5\text{O}_3^-(\text{aq}) + \text{C}_2\text{H}_4\text{O}_3(\text{aq})$
 B. $\text{C}_7\text{H}_6\text{O}_3(\text{aq}) + \text{C}_3\text{H}_5\text{O}_3^-(\text{aq}) \rightleftharpoons \text{C}_7\text{H}_5\text{O}_3^-(\text{aq}) + \text{C}_3\text{H}_6\text{O}_3(\text{aq})$
 C. $\text{C}_7\text{H}_6\text{O}_3(\text{aq}) + \text{C}_2\text{H}_2\text{ClO}_2^-(\text{aq}) \rightleftharpoons \text{C}_2\text{H}_3\text{O}_3^-(\text{aq}) + \text{C}_2\text{H}_3\text{ClO}_2(\text{aq})$
 D. $\text{C}_2\text{H}_4\text{O}_3(\text{aq}) + \text{C}_2\text{H}_2\text{ClO}_2^-(\text{aq}) \rightleftharpoons \text{C}_7\text{H}_5\text{O}_3^-(\text{aq}) + \text{C}_2\text{H}_3\text{ClO}_2(\text{aq})$

Use the following information to answer the next question.

A student was asked to rank the relative strength of the following four acids.

1. Formic Acid ($\text{HCOOH}(\text{aq})$)
2. Hydrazoic acid ($\text{HN}_3(\text{aq})$)
3. Hypobromous acid ($\text{HOBr}(\text{aq})$)
4. Nitrous acid ($\text{HNO}_2(\text{aq})$)

The student was given the following evidence.



Numerical Response

49. Based on the reaction evidence, the four acids, ranked from strongest to weakest, are _____, _____, _____, and _____.
50. Vinegar ($\text{CH}_3\text{COOH}(\text{aq})$) and baking soda ($\text{NaHCO}_3(\text{s})$) are added to recipes to produce baked products with light, fluffy textures. The net ionic equation for the reaction that occurs is
- A. $\text{H}_3\text{O}^+(\text{aq}) + \text{HCO}_3^-(\text{aq}) \rightleftharpoons \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$
 - B. $\text{HCO}_3^-(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq}) \rightleftharpoons \text{CH}_3\text{COOH}(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$
 - C. $\text{CH}_3\text{COOH}(\text{aq}) + \text{HCO}_3^-(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
 - D. $\text{CH}_3\text{COOH}(\text{aq}) + \text{NaHCO}_3(\text{aq}) \rightleftharpoons \text{NaCH}_3\text{COO}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
51. A reaction favouring the reactants in which $\text{HCO}_3^-(\text{aq})$ acts as an acid is
- A. $\text{HCO}_3^-(\text{aq}) + \text{PO}_4^{3-}(\text{aq}) \rightleftharpoons \text{HPO}_4^{2-}(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$
 - B. $\text{HCO}_3^-(\text{aq}) + \text{HPO}_4^{2-}(\text{aq}) \rightleftharpoons \text{H}_2\text{PO}_4^-(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$
 - C. $\text{HCO}_3^-(\text{aq}) + \text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
 - D. $\text{HCO}_3^-(\text{aq}) + \text{HSO}_4^-(\text{aq}) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$

Use the following information to answer the next question.

Chemical Species

1. $\text{HA}^{3-}(\text{aq})$
2. $\text{H}_3\text{A}^-(\text{aq})$
3. $\text{H}_2\text{A}^{2-}(\text{aq})$
4. $\text{H}_4\text{A}(\text{aq})$

Numerical Response

52. As a solution of $\text{NaOH}(\text{aq})$ is continuously added to the acid $\text{H}_4\text{A}(\text{aq})$, a sequence of quantitative reactions occur. The order in which the species listed above would react is _____, _____, _____, and _____.
53. Which of the following equations represents an equilibrium system that favours the products?
- A. $\text{H}_2\text{S}(\text{aq}) + \text{F}^-(\text{aq}) \rightleftharpoons \text{HS}^-(\text{aq}) + \text{HF}(\text{aq})$
 - B. $\text{HF}(\text{aq}) + \text{NO}_2^-(\text{aq}) \rightleftharpoons \text{F}^-(\text{aq}) + \text{HNO}_2(\text{aq})$
 - C. $\text{HSO}_3^-(\text{aq}) + \text{HCN}(\text{aq}) \rightleftharpoons \text{H}_2\text{SO}_3(\text{aq}) + \text{CN}^-(\text{aq})$
 - D. $\text{HOCl}(\text{aq}) + \text{HCO}_3^-(\text{aq}) \rightleftharpoons \text{OCl}^-(\text{aq}) + \text{H}_2\text{CO}_3(\text{aq})$
54. Pickling vinegar has a pH of 2.37. When 3 drops of bromocresol green and 3 drops of phenolphthalein are added to a sample of this vinegar, the resulting color of the solution is
- A. blue
 - B. green
 - C. yellow
 - D. purple
55. Which of the following solutions could act as a buffer system?
- A. $\text{KH}_2\text{PO}_4(\text{aq}) / \text{H}_3\text{PO}_4(\text{aq})$
 - B. $\text{KCl}(\text{aq}) / \text{HCl}(\text{aq})$
 - C. $\text{KClO}_4(\text{aq}) / \text{HClO}_4(\text{aq})$
 - D. $\text{H}_3\text{O}^+(\text{aq}) / \text{OH}^-(\text{aq})$

56. A solution was tested and found to have a pOH of 3.2. This solution would **most likely** be a proton i , and turn ii .

The statement above is completed by the information in row

Row	i	ii
A.	acceptor	thymolphthalein blue
B.	donor	thymolphthalein blue
C.	acceptor	thymol blue yellow
D.	donor	thymol blue yellow

Use the following information to answer the next question.

The labels came off four cleaning solution containers found under a kitchen sink. Each of the cleaning solutions was tested with two available indicators and the following results were recorded.

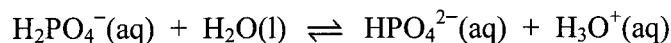
Cleaning Solution	Bromothymol blue	Phenolphthalein
1	blue	pink
2	blue	colourless
3	green	colourless
4	blue	light pink

Numerical Response

57. Listed in order from lowest to highest pH, the cleaning solutions are, respectively _____, _____, _____, and _____.

Use the following information to answer the next question.

The following equilibrium system maintains a constant pH in the intracellular fluid in the body.



Terms Associated with Acid-Base Equilibrium Systems

- 1** Buffer
- 2** Amphiprotic
- 3** Polyprotic acid
- 4** Polyprotic base
- 5** Conjugate acid-base pair

- 58.** Which of the terms numbered above can be used to describe the reaction or the species in the reaction in the equilibrium system above?
- A.** 1 and 5 only
 - B.** 2 and 3 only
 - C.** 2, 4, and 5 only
 - D.** 1, 2, 3, 4, and 5
- 59.** Blood maintain a nearly constant pH because it contains
- A.** hemoglobin that maintains the oxygen levels in the blood
 - B.** catalysts (enzymes) that control the equilibrium in the blood
 - C.** sodium ions and chloride ions that keep the pH of the blood at 7
 - D.** buffers that regulate the hydronium ion concentration in the blood

Use the following information to answer the next question.

A sample of rainwater is poured into five test tubes. A different indicator is added to each test tube. Four of the observations are recorded in the table below.

Indicator	Colour
Methyl red	Yellow
Phenol red	Yellow
Bromocresol green	Blue
Phenolphthalein	Colourless
Bromothymol blue	?

60. The pH of the rainwater and the predicted color of the sample containing bromothymol blue are
- A. 6.0 and blue
 - B. 7.6 and blue
 - C. 6.0 and yellow
 - D. 7.6 and yellow
61. Which of the following rows identifies the equilibrium equation and solution colour that would occur when a few drops of phenol red indicator, $\text{HPr}(\text{aq})/\text{Pr}^-(\text{aq})$, are added to a 100.00 mL sample of 0.50 mol/L nitrous acid?

Row	Equilibrium Equations	Solution Colour
A.	$\text{HNO}_2(\text{aq}) + \text{Pr}^-(\text{aq}) \rightleftharpoons \text{NO}_2^-(\text{aq}) + \text{HPr}(\text{aq})$	Yellow
B.	$\text{HNO}_2(\text{aq}) + \text{Pr}^-(\text{aq}) \rightleftharpoons \text{NO}_2^-(\text{aq}) + \text{HPr}(\text{aq})$	Red
C.	$\text{HNO}_2(\text{aq}) + \text{HPr}(\text{aq}) \rightleftharpoons \text{NO}_2^-(\text{aq}) + \text{H}_2\text{Pr}^+(\text{aq})$	Yellow
D.	$\text{HNO}_2(\text{aq}) + \text{HPr}(\text{aq}) \rightleftharpoons \text{NO}_2^-(\text{aq}) + \text{H}_2\text{Pr}^+(\text{aq})$	Red

Use the following information to answer the next **two** questions.

Selected Acids and Bases

I	$\text{H}_2\text{O}(\text{l})$
II	$\text{OH}^-(\text{aq})$
III	$\text{OCl}^-(\text{aq})$
IV	$\text{HC}_6\text{H}_5\text{O}_6^-(\text{aq})$
V	$\text{HOCCCOO}^-(\text{aq})$
VI	$\text{HOCCCOOH}(\text{aq})$

62. The amphiprotic species listed above are
- A. I, II, and V
B. I, III, and VI
C. I, IV, and V
D. I, V, and VI
63. The two species listed above that could be combined to prepare a buffer solution are
- A. I and III
B. III and IV
C. III and VI
D. V and VI

Use the following information to answer the next question.

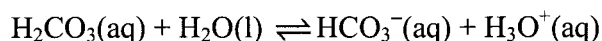
A beaker contains 100 mL of a buffer comprised of 0.10 mol/L $\text{NaCH}_3\text{COO}(\text{aq})$ and 0.10 mol/L $\text{CH}_3\text{COOH}(\text{aq})$.

64. A small amount of $\text{NaOH}(\text{aq})$ is added to this buffer. The net ionic equation that represents what occurs is
- A. $\text{CH}_3\text{COOH}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
B. $\text{CH}_3\text{COO}^-(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{CH}_3\text{COOH}(\text{aq}) + \text{O}_2^-(\text{aq})$
C. $\text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
D. $\text{CH}_3\text{COOH}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCH}_3\text{COO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

65. The main buffer solution of plasma and tissue fluid found in our bodies is $\text{H}_2\text{CO}_3(\text{aq}) / \text{HCO}_3^-(\text{aq})$. When excess hydronium ions enter our blood, the equation that represents the reaction that occurs is
- A. $\text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow 2 \text{H}_2\text{O}(\text{l})$
 - B. $\text{H}_2\text{CO}_3(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 - C. $\text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$
 - D. $\text{H}_3\text{O}^+(\text{aq}) + \text{HCO}_3^-(\text{aq}) \rightarrow \text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l})$
66. Which of the following equimolar solutions could act as a buffer system?
- A. $\text{KCl}(\text{aq})/\text{HCl}(\text{aq})$
 - B. $\text{KNO}_3(\text{aq})/\text{HNO}_3(\text{aq})$
 - C. $\text{KClO}_4(\text{aq})/\text{HClO}_4(\text{aq})$
 - D. $\text{KH}_2\text{PO}_4(\text{aq}) / \text{H}_3\text{PO}_4(\text{aq})$
67. Which of the following pairs of chemical compounds can act as a buffer system?
- A. $\text{HCl}(\text{aq})$ and $\text{NaOH}(\text{aq})$
 - B. $\text{HCl}(\text{aq})$ and $\text{H}_2\text{SO}_4(\text{aq})$
 - C. $\text{CH}_3\text{COOH}(\text{aq})$ and $\text{HCl}(\text{aq})$
 - D. $\text{CH}_3\text{COOH}(\text{aq})$ and $\text{NaCH}_3\text{COO}(\text{aq})$
68. The concentration of $\text{H}_3\text{O}^+(\text{aq})$ in a $0.040 \text{ mol/L } \text{H}_2\text{S}(\text{aq})$ solution is
- A. $3.0 \times 10^{-4} \text{ mol/L}$
 - B. $6.0 \times 10^{-5} \text{ mol/L}$
 - C. $2.2 \times 10^{-6} \text{ mol/L}$
 - D. $3.6 \times 10^{-9} \text{ mol/L}$
69. Hydrofluoric acid is a weak acid but can be very corrosive. The K_b of the conjugate base, $\text{F}^-(\text{aq})$, is
- A. 1.6×10^{17}
 - B. 6.6×10^{-4}
 - C. 1.6×10^{-11}
 - D. 1.0×10^{-14}

Use the following information to answer the next question.

A 100.0 mL sample of a 0.100 mol/L solution of $\text{H}_2\text{CO}_3(\text{aq})$ is combined with a 100.0 mL sample of a 0.100 mol/L solution of $\text{NaHCO}_3(\text{aq})$ and the following equilibrium system is established.



70. Which of the following rows identifies the direction of the shift in equilibrium and the effect on the pH of the system when a small quantity of hydrochloric acid is added to the equilibrium system above?

Row	Direction of Shift	Effect on pH
A.	toward the products	no change
B.	toward the products	decrease
C.	toward the reactants	no change
D.	toward the reactants	decrease

71. Ammonia, $\text{NH}_3(\text{aq})$, can be described as a i base, and the concentration of hydroxide ions in a solution of ammonia with a pH of 8.10 is ii .

The statement above is completed by the information in row

Row	i	ii
A.	weak	$1.3 \times 10^{-6} \text{ mol/L}$
B.	weak	$7.9 \times 10^{-9} \text{ mol/L}$
C.	strong	$1.3 \times 10^{-6} \text{ mol/L}$
D.	strong	$7.9 \times 10^{-9} \text{ mol/L}$

72. A cleaning agent has a pH of 1 and a carbonated beverage has a pH of 5. The cleaning agent is more acidic than the carbonated beverage by a factor of
- A. 10
B. 100
C. 1 000
D. 10 000

Numerical Response

73. The pH of a 0.13 mol/L solution of propanoic acid is _____.

Use the following information to answer the next **three** questions.

Sodium azide, which is found in automobile airbags, reacts readily with acids to form the highly toxic and explosive hydroazoic acid, $\text{HN}_3(\text{aq})$. The K_a for hydroazoic acid is 1.9×10^{-5} .

74. The K_a expression for hydroazoic acid is

A. $K_a = \frac{[\text{HN}_3(\text{aq})]}{[\text{H}_3\text{O}^+(\text{aq})][\text{N}_3^-(\text{aq})]}$

B. $K_a = \frac{[\text{H}_3\text{O}^+(\text{aq})][\text{N}_3^-(\text{aq})]}{[\text{HN}_3(\text{aq})]}$

C. $K_a = \frac{[\text{HN}_3(\text{aq})]^3}{[\text{H}_3\text{O}^+(\text{aq})][\text{N}_3^-(\text{aq})]^3}$

D. $K_a = \frac{[\text{H}_3\text{O}^+(\text{aq})][\text{N}_3^-(\text{aq})]^3}{[\text{HN}_3(\text{aq})]^3}$

Numerical Response

75. The pH of a 0.28 mol/L $\text{HN}_3(\text{aq})$ solution is _____.

76. In terms of equilibrium concentrations, in a solution of hydroazoic acid,

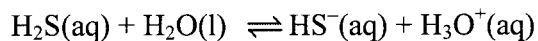
- A. $[\text{HN}_3(\text{aq})] < [\text{N}_3^-(\text{aq})]$
B. $[\text{HN}_3(\text{aq})] > [\text{H}_3\text{O}^+(\text{aq})]$
C. $[\text{HN}_3(\text{aq})] > [\text{H}_2\text{O}(\text{l})]$
D. $[\text{HN}_3(\text{aq})] = [\text{H}_3\text{O}^+(\text{aq})]$

77. Sour pickles have a pH of about 3.00. The concentration of hydroxide ions in sour pickle brine is

- A. 1.0×10^{-11} mol/L
- B. 3.0×10^{-11} mol/L
- C. 1.0×10^{-3} mol/L
- D. 3.0×10^{-3} mol/L

Use the following information to answer the next two questions.

If $\text{H}_2\text{S}(\text{g})$ is released into the atmosphere, it dissolves in atmospheric water to form hydrosulfuric acid, $\text{H}_2\text{S}(\text{aq})$. The ionization of $\text{H}_2\text{S}(\text{aq})$ can be represented by the following equilibrium equation.



78. The K_a expression for this ionization is

- A. $K_a = \frac{[\text{H}_2\text{S}(\text{aq})]}{[\text{HS}^-(\text{aq})][\text{H}_3\text{O}^+(\text{aq})]}$
- B. $K_a = \frac{[\text{H}_2\text{S}(\text{aq})][\text{H}_2\text{O}(\text{l})]}{[\text{HS}^-(\text{aq})][\text{H}_3\text{O}^+(\text{aq})]}$
- C. $K_a = \frac{[\text{HS}^-(\text{aq})][\text{H}_3\text{O}^+(\text{aq})]}{[\text{H}_2\text{S}(\text{aq})][\text{H}_2\text{O}(\text{l})]}$
- D. $K_a = \frac{[\text{HS}^-(\text{aq})][\text{H}_3\text{O}^+(\text{aq})]}{[\text{H}_2\text{S}(\text{aq})]}$

79. If the concentration of a sample of aqueous hydrosulfuric acid is 0.050 mol/L, then the hydronium ion concentration is i and the pOH of the solution is ii .

The statement above is completed by the information in row

Row	i	ii
A.	5.0×10^{-2} mol/L	1.30
B.	5.0×10^{-2} mol/L	9.82
C.	6.7×10^{-5} mol/L	1.30
D.	6.7×10^{-5} mol/L	9.82

Use the following information to answer the next question.

Solutions of carbolic acid, commonly known as phenol ($\text{HC}_6\text{H}_5\text{O}(\text{aq})$) are widely used as disinfectants. One such solution has a concentration of 6.44×10^{-2} mol/L and a pH of 5.60. Carboxylic acid dissociates in water according to the equation



80. The K_a expression for this reaction is

A. $K_c = \frac{[\text{C}_6\text{H}_5\text{O}^-(\text{aq})][\text{H}_3\text{O}^+(\text{aq})]}{[\text{HC}_6\text{H}_5\text{O}(\text{aq})][\text{H}_2\text{O}(\text{l})]}$

B. $K_c = \frac{[\text{HC}_6\text{H}_5\text{O}(\text{aq})][\text{H}_2\text{O}(\text{l})]}{[\text{C}_6\text{H}_5\text{O}^-(\text{aq})][\text{H}_3\text{O}^+(\text{aq})]}$

C. $K_c = \frac{[\text{C}_6\text{H}_5\text{O}^-(\text{aq})][\text{H}_3\text{O}^+(\text{aq})]}{[\text{HC}_6\text{H}_5\text{O}(\text{aq})]}$

D. $K_c = \frac{[\text{HC}_6\text{H}_5\text{O}(\text{aq})]}{[\text{C}_6\text{H}_5\text{O}^-(\text{aq})][\text{H}_3\text{O}^+(\text{aq})]}$

Use the following information to answer the next question.

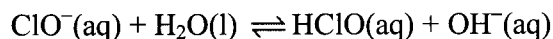
The Cancarb plant in Medicine Hat produces carbon, C(s), from methane gas. To cool the carbon in the production line, it is sprayed with water. In this process, the water becomes increasingly basic. The $[\text{OH}^-(\text{aq})]$ of the resulting solution is $3.2 \times 10^{-4} \text{ mol/L}$.

81. The pH of the solution is

- A. 0.50
- B. 3.49
- ☒ C. 10.51
- D. 13.50

Use the following information to answer the next question.

Hot tub owners can control disease causing bacteria and algae by adding solid sodium hypochlorite pellets ($\text{NaClO}(\text{s})$) to the water. This results in the formation of $\text{HClO}(\text{aq})$ as represented by the equilibrium equation



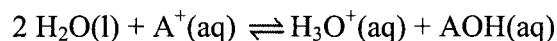
Undissociated $\text{HClO}(\text{aq})$ effectively kills bacteria and algae. A pH of 7.40 is considered ideal for a hot tub.

82. Ideally the water in a hot tub should have a hydronium ion concentration of

- A. $4.0 \times 10^{-8} \text{ mol/L}$ and is basic
- B. $2.5 \times 10^{-7} \text{ mol/L}$ and is basic
- C. $4.0 \times 10^{-8} \text{ mol/L}$ and is acidic
- D. $2.5 \times 10^{-7} \text{ mol/L}$ and is acidic

Use the following information to answer the next question.

Anthocyanins (represented as $A^+(aq)$) are responsible for the red colour of wine. Anthocyanins can react with water, as described by the following equation



The $AOH(aq)$ form is aqueous. In a particular sample of white wine, the $[H_3O^+(aq)]$ at equilibrium is $8.5 \times 10^{-4} \text{ mol/L}$.

83. If the initial concentration of anthocyanins, prior to equilibrium, was $1.62 \times 10^{-3} \text{ mol/L}$, the K_a for anthocyanins is
- A. 2.24×10^3
B. 1.07×10^1
C. 9.38×10^{-4}
D. 4.46×10^{-4}

Numerical Response

84. When 0.10 mol of $NH_4NO_3(s)$ is dissolved in water to produce 1.0 L of solution, the pH of the solution is _____.
85. If the pH of a solution changes from 2 to 4, then the hydronium ion concentration
- A. is halved
B. is doubled
C. increases by a factor of 100
D. decreases by a factor of 100

Numerical Response

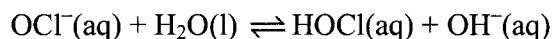
86. The hydroxide ion concentration in a solution with a pH of 3.50, expressed in scientific notation, is $a.b \times 10^{-cd} \text{ mol/L}$. The values of a , b , c , and d are _____, _____, _____, and _____.

Numerical Response

87. The K_b for the ethanoate ion, $CH_3COO^-(aq)$, expressed in scientific notation, is $a.b \times 10^{-cd}$. The values of a , b , c , and d are _____, _____, _____, and _____.

Use the following information to answer the next question.

Household bleach contains sodium hypochlorite, which is a weak base with a K_b of 3.4×10^{-7} . The ionization of sodium hypochlorite can be represented by the following equilibrium equation.



88. The pOH of a 0.012 5 mol/L solution of sodium hypochlorite is
- A. 1.90
B. 4.19
C. 8.37
D. 9.81

Use the following information to answer the next question.

Sodium fluoroethanoate, $\text{NaCH}_2\text{FCOO}(\text{aq})$, is a potent metabolic poison that can be used to kill rodents. The conjugate acid of the fluoroethanoate ion is fluoroethanoic acid, $\text{CH}_2\text{FCOOH}(\text{aq})$. For this acid, $K_a = 2.6 \times 10^{-3}$.

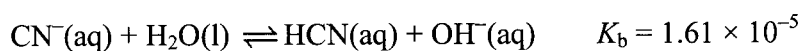
89. The value of K_b of $\text{CH}_2\text{FCOO}^-(\text{aq})$ is *i* , and $\text{CH}_2\text{FCOO}^-(\text{aq})$ is a stronger base than *ii* .

The statement above is completed by the information in row

Row	<i>i</i>	<i>ii</i>
A.	3.8×10^{-12}	$\text{H}_2\text{PO}_4^-(\text{aq})$
B.	3.8×10^{-12}	$\text{PO}_4^{3-}(\text{aq})$
C.	2.6×10^{-3}	$\text{H}_2\text{PO}_4^-(\text{aq})$
D.	2.6×10^{-3}	$\text{PO}_4^{3-}(\text{aq})$

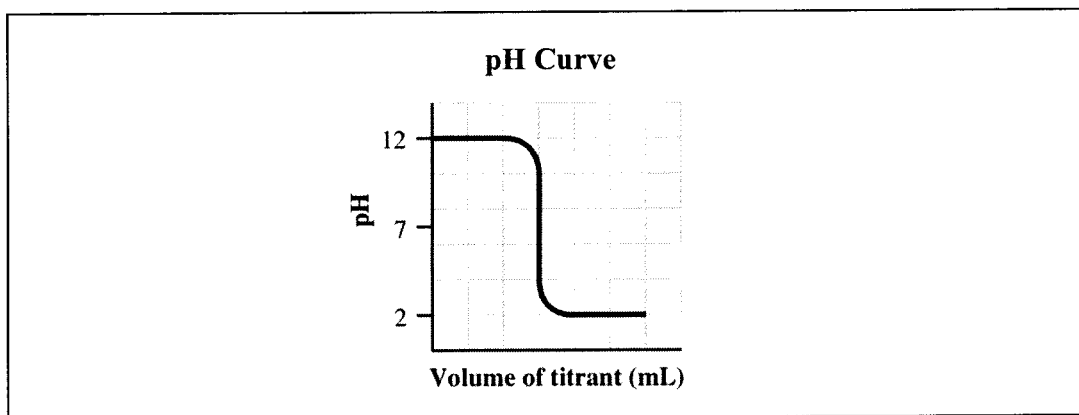
90. A brand of window-cleaning fluid is composed of an ammonia solution, water, a colouring agent, and a very small quantity of soap. It has a 0.15 mol/L concentration of $\text{NH}_3(\text{aq})$. The pH of this fluid is
- A. 2.79
 - B. 5.04
 - C. 8.96
 - D. 11.21

Use the following information to answer the next question.



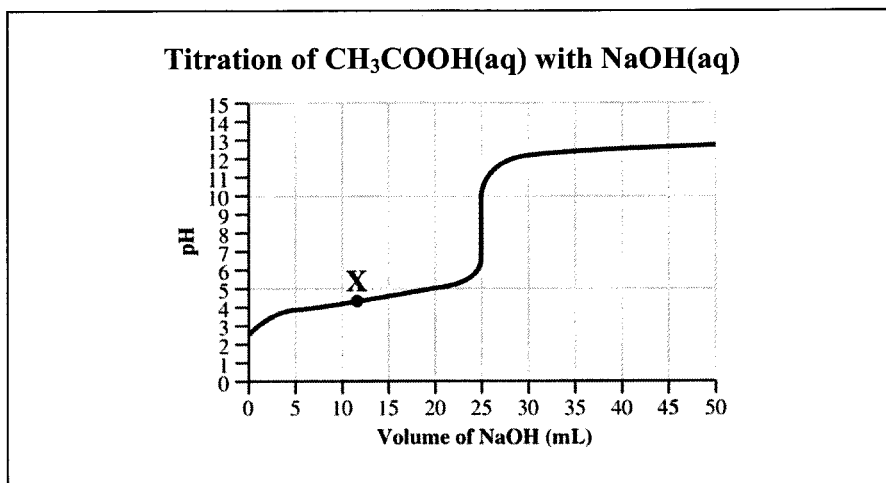
91. If the pH of a solution of $\text{NaCN}(\text{aq})$ is 8.710, then the equilibrium concentration of $\text{CN}^-(\text{aq})$, expressed in scientific notation, is $a.bc \times 10^{-d}$ mol/L. The values of *a*, *b*, *c*, and *d* are _____, _____, _____, and _____.

Use the following information to answer the next question.



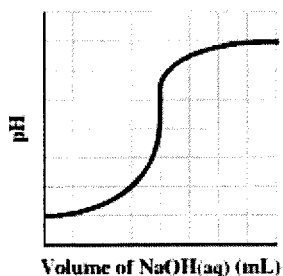
92. The graph represents the titration of
- A. hydrochloric acid with ammonia
 - B. acetic acid with sodium hydroxide
 - C. potassium hydroxide with oxalic acid
 - D. sodium hydroxide with hydrochloric acid

Use the following information to answer the next question.

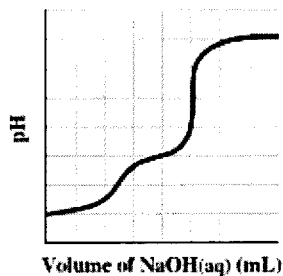


93. If the titration is stopped at X, the solution is resistant to a change in pH if a strong base or a strong acid is added to it. This is due to the fact that, at X, the solution contains large amounts of
- A. $\text{H}_2\text{O}(\text{l})$ and $\text{OH}^-(\text{aq})$
B. $\text{CH}_3\text{COO}^-(\text{aq})$ and $\text{H}_3\text{O}^+(\text{aq})$
C. $\text{CH}_3\text{COOH}(\text{aq})$ and $\text{OH}^-(\text{aq})$
D. $\text{CH}_3\text{COOH}(\text{aq})$ and $\text{CH}_3\text{COO}^-(\text{aq})$
94. When oxalic acid is titrated with $\text{NaOH}(\text{aq})$, the titration curve that would be predicted is

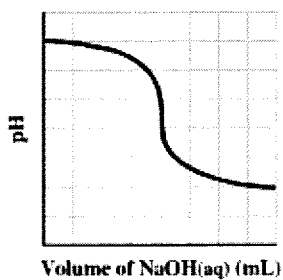
A.



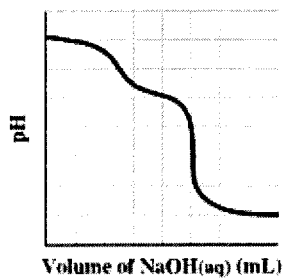
B.



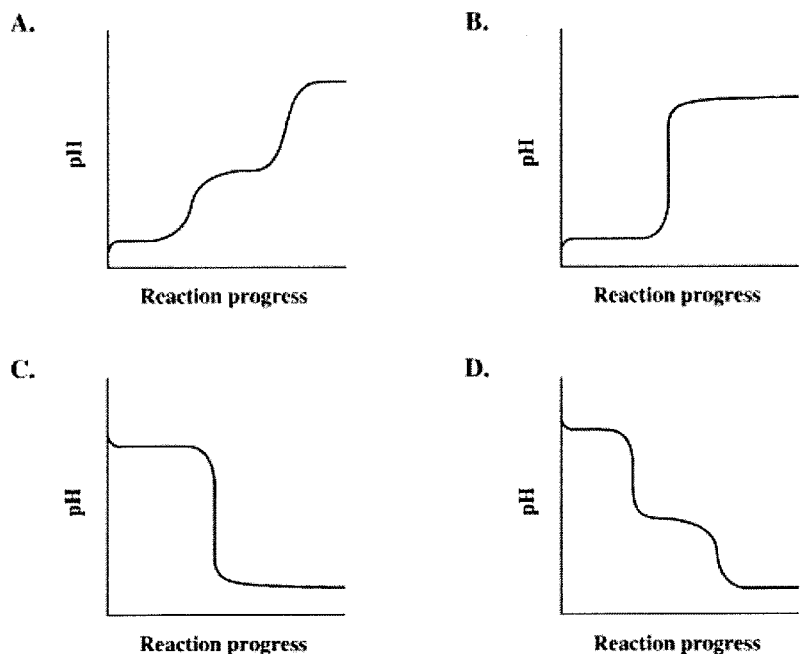
C.



D.

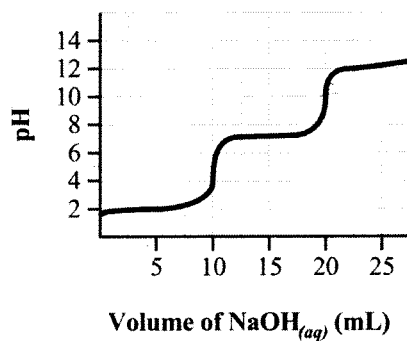


95. Which of the following pH curves represents the titration of an acid with a strong base, if one intermediate amphoteric substance is produced?



Use the following information to answer the next question.

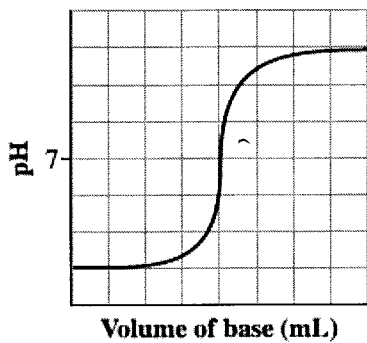
A student sketched a titration curve based on data collected during a reaction between a 0.050 mol/L NaOH(aq) titrant and a 25.0 mL sample of a soft drink.



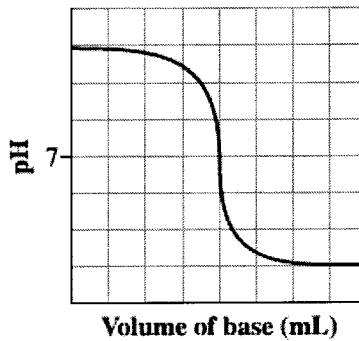
96. The most suitable indicator to identify the equivalence point of the second reaction is
- A. methyl red
 - B. methyl orange
 - C. phenolphthalein
 - D. bromothymol blue

97. Which of the following diagrams represents the titration of a polyprotic acid with a strong base?

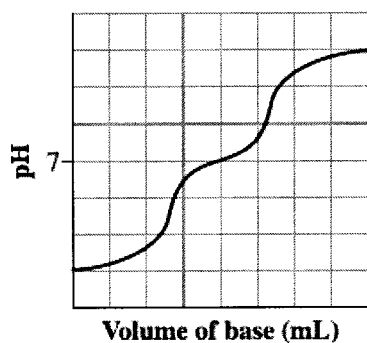
A.



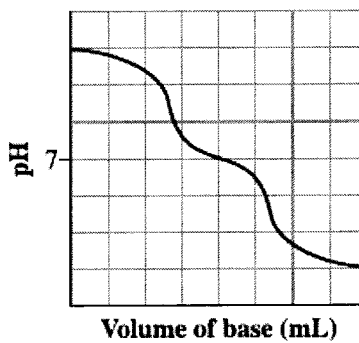
B.



C.



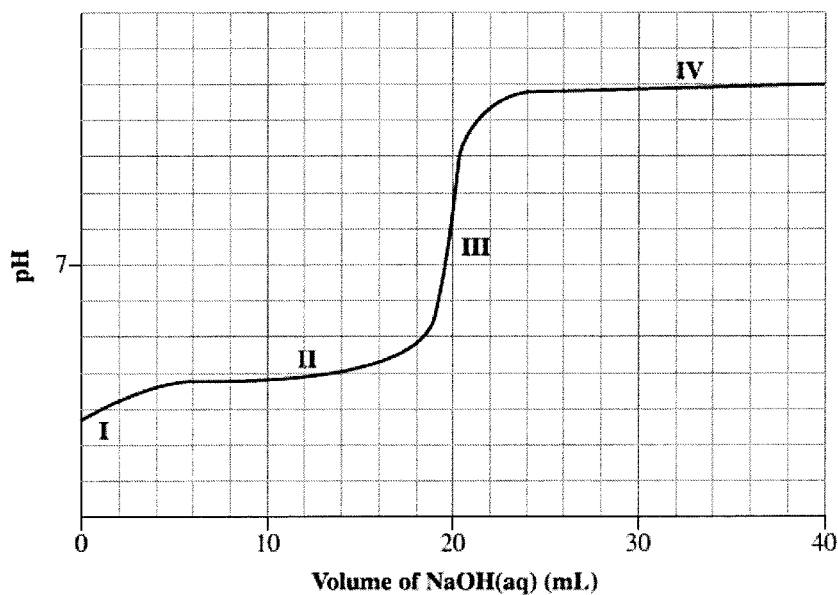
D.



Use the following information to answer the next question.

A student titrated an unknown sample of an aqueous acid with aqueous sodium hydroxide. The student graphed the data as shown in the diagram below.

Graph of Titration Data



98. The graph of the student's titration data shows the titration of a i acid with a strong base, and on the graph, buffering occurs at ii.

The statement above is completed by the information in row

Row	i	ii
A.	strong	region II only
B.	strong	regions II and IV
C.	weak	region II only
D.	weak	regions II and IV

Unit D: Chemical Equilibrium Focusing on Acid-Base Systems – Answers

- | | | | |
|----------|----------|----------|----------|
| 1. A | 31. D | 61. A | 91. 1636 |
| 2. A | 32. 4123 | 62. C | 92. D |
| 3. A | 33. 3578 | 63. D | 93. D |
| 4. C | 34. B | 64. C | 94. B |
| 5. D | 35. A | 65. D | 95. A |
| 6. 60.0 | 36. C | 66. D | 96. C |
| 7. A | 37. D | 67. D | 97. C |
| 8. B | 38. D | 68. B | 98. C |
| 9. B | 39. C | 69. C | |
| 10. 43.7 | 40. A | 70. C | |
| 11. B | 41. D | 71. A | |
| 12. 9.93 | 42. C | 72. D | |
| 13. D | 43. C | 73. 2.89 | |
| 14. A | 44. D | 74. B | |
| 15. D | 45. 1423 | 75. 2.64 | |
| 16. C | 46. A | 76. B | |
| 17. 87.7 | 47. B | 77. A | |
| 18. D | 48. B | 78. D | |
| 19. A | 49. 4123 | 79. D | |
| 20. C | 50. C | 80. C | |
| 21. 3142 | 51. B | 81. C | |
| 22. B | 52. 4231 | 82. C | |
| 23. 1367 | 53. B | 83. C | |
| 24. B | 54. C | 84. 5.13 | |
| 25. C | 55. A | 85. D | |
| 26. A | 56. A | 86. 3211 | |
| 27. 1358 | 57. 3241 | 87. 5610 | |
| 28. C | 58. D | 88. B | |
| 29. C | 59. D | 89. A | |
| 30. A | 60. C | 90. D | |

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